

University of Dhaka
Affiliated Engineering Colleges/Institute
Department of Computer Science and Engineering
2nd Year 1st Semester B.Sc. Final Examination, 2023
Course Code: CSE-2103
Course Name: Digital Electronics and Pulse Techniques

Time: 3 Hours

Full Marks: 5x14=70

Answer any 5 (Five) set of the following Questions

1. a) Show that Transistor acts as a switch. 3
b) Design NAND, NOR gate using CMOS logic. 4
c) Design a two input NOR gate using TTL circuit and explain its operation. 4
d) Define the parameters: i) Noise Margin ii) Propagation Delay iii) Fan-in 3
2. a) Describe the order and draw the general structure of PLA circuit. 4
b) Write the steps for implementing a PLA. 3
c) A combinational logic is defined by the following function:
 $F1(A,B,C) = \sum m(3,5,6,7)$
 $F2(A,B,C) = \sum m(0,2,4,7)$
Design the circuit with PLA having 3 inputs, 4 product terms and 2 outputs. 4
d) Explain CMOS acts as inverter. 3
3. a) Differentiate between latch and flip-flop. 3
b) What is the problem associated with a SR flip-flop? How does the issue can be resolved by D-flip flop. 4
c) What is a clock signal? Mention the difference between the edge triggered clock and level triggered clock using a waveform. 3
d) Explain the role of preset and clear input in a J-K flip flop, where a square wave is used as clock and the both inputs of flop-flop are high. 4
4. a) What is toggle mode in J-K flip-flop? Mention the different application of the toggle mode of J-K flip flop. 3
b) Suppose you have a fixed frequency generator of 1KHz. You need to use a 100 Hz frequency in a circuit. How will you accomplish this task? Explain using necessary diagram. 4
c) What is asynchronous counter? What are the differences between the synchronous and asynchronous counter. 3
d) Design a Mod 10 Synchronous binary counter with timing diagram for one cycle. 4
5. a) Mention the characteristics of an operational amplifier. Draw the circuit symbol and identify different important points. 4
b) Explain with necessary circuit diagram how could an op-amp be used as a comparator. 3
c) Explain the working principle of an astable multivibrator using op-amp. Also, illustrate the input and output wave shape of the circuit. 7
6. a) Define the following terms: 4
i) Resolution (ii) Step size (iii) Full-scale output (iv) Full-scale error.
b) A 10-bit DAC has a step size of 10mV. Determine the full-scale output voltage and the percentage resolution. 2
c) Describe the basic operation of the digital-ramp ADC. 4
d) Explain the memory Read and Write operation. 4
7. a) What is meant by LED and LCD. 2
b) Implement Sample and Hold (S/H) circuit by using Operational Amplifier. 4
c) Describe binary weighted resistor Digital to Analog Converter circuit. 4
- d) Find the output of the following circuits for sinusoidal input. Also explain the operation of the circuits. 4

